

# CPSC 3100 HW 3

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## A.1

As you can see, the top 10 candidates were all males when sorted by the standard f1 scores. For full sorted dataset see applicants1.csv.

| ID  | Gender | University | Skills_Test | Overtime | Experience | F1 Score |
|-----|--------|------------|-------------|----------|------------|----------|
| 133 | M      | 10         | 10          | 8        | 10         | 9.50     |
| 42  | M      | 10         | 9           | 10       | 9          | 9.50     |
| 118 | M      | 9          | 10          | 9        | 9          | 9.25     |
| 7   | M      | 8          | 10          | 9        | 9          | 9.00     |
| 17  | M      | 9          | 9           | 10       | 8          | 9.00     |
| 14  | M      | 7          | 9           | 10       | 10         | 9.00     |
| 9   | M      | 9          | 10          | 9        | 8          | 9.00     |
| 151 | M      | 10         | 9           | 8        | 9          | 9.00     |
| 82  | M      | 8          | 9           | 9        | 10         | 9.00     |
| 79  | M      | 8          | 9           | 10       | 9          | 9.00     |

Table 1: Top candidates by F1 score

## A.2

### A.2.1

This classifier I defined for f2 was

$$f_2(x) = 0.7x_1 + 0.7x_2 + 0.05x_3 + 0.55x_4$$

where  $x_1, x_2, x_3, x_4$  represent university, skills test, overtime, and experience, respectively.

This classifier reduces the weight of overtime and experience, which are higher on average for male applicants, and increases the weight of university and skills.

As a result, applicants are ranked more based on academic quality and skills, which allows at least three women to appear in the top 10 applicants.

| Gender | University | Skills_Test | Overtime | Experience |
|--------|------------|-------------|----------|------------|
| F      | 5.522222   | 5.277778    | 5.033333 | 6.166667   |
| M      | 5.881818   | 5.563636    | 7.145455 | 7.745455   |

Table 2: Average values by gender, printed from my code

### A.2.2

For any applicants  $x$  and  $y$ ,

$$f_2(x) - f_2(y) = 0.7(x_1 - y_1) + 0.7(x_2 - y_2) + 0.05(x_3 - y_3) + 0.55(x_4 - y_4)$$

Taking absolute values,

$$|f_2(x) - f_2(y)| \leq 0.7|x_1 - y_1| + 0.7|x_2 - y_2| + 0.05|x_3 - y_3| + 0.55|x_4 - y_4|$$

Since the sum of the weights is

$$0.7 + 0.7 + 0.05 + 0.55 = 2,$$

we have

$$|f_2(x) - f_2(y)| \leq 2 \cdot \frac{1}{4} \sum_{i=1}^4 |x_i - y_i|$$

Dividing both sides by 9,

$$D(f_2(x), f_2(y)) = \frac{1}{9} |f_2(x) - f_2(y)| \leq 2 \cdot \frac{1}{36} \sum_{i=1}^4 |x_i - y_i| = 2d(x, y)$$

Therefore,

$$D(f_2(x), f_2(y)) \leq 2d(x, y)$$

for all applicants  $x$  and  $y$ .

### A.2.3

I created a new dataset `applicants2.csv` by adding a column labeled `f2 score` to `applicants1.csv`. For each applicant  $x$ , I computed  $f_2(x)$  using the above definition and inserted the result into the new column.

The dataset was then sorted in descending order by `f2 score`. The resulting ranking includes at least three women in the top 10 applicants.

## A.3

Using  $f_2$  is better than  $f_1$  because  $f_1$  results in the top 10 applicants being all men, even though there are qualified women in the dataset. This happens

because features such as overtime and experience, which are weighted equally in  $f_1$ , tend to be higher on average for male applicants as men are traditionally expected to be in the workplace so women would have less experience. Unfair hiring advantages in other companies may affect women’s experience levels, and needing to be at home tending to children etc, can affect overtime hours.

The classifier  $f_2$  adjusts the weights by reducing the influence of overtime and experience and increasing the importance of university and skills, which are arguably more important in qualifying a candidate. This change allows applicants to be evaluated more based on academic quality and demonstrated ability, which brings more qualified women into the top 10. At the same time,  $f_2$  still satisfies the relaxed fairness condition  $D(f_2(x), f_2(y)) \leq 2d(x, y)$ , so it maintains a bounded notion of fairness while improving representation.

## A.4

One argument against using  $f_2$  is that the inequality between men and women in fields like overtime may simply and truly be unrelated to historical inequalities between men and women. For example, in the dataset, we already see that women and men are quite equal on average when it comes to skills and university degree. That means that the historical inequality between the two genders may not be that big of an issue. So perhaps the experience difference in women and men are not correlated in the dataset, and women in this hiring cycle just tend to have less experience. Also, if this job is attracting the same age group of applicants (recent MS degrees) then big reasons for less overtime for women, such as childcare etc., is probably not the reason that the two genders are so disparate. (also with men becoming more and more involved in the home as of recent years, this further implicates that overtime in applicants may just be more appararent in men this hiring cycle.)

In addition, modifying the classifier to influence outcomes can be seen as weakening the original idea of treating similar individuals similarly. Instead of changing the scoring function, a better approach might be to keep  $f_1$  and apply a separate policy when selecting candidates for interviews, such as ensuring diversity at a later stage. This would preserve the original definition of fairness while still addressing representation concerns.

## Part B

### B.1

One rule that a social-media company could adopt is that all content identified as hate speech or explicit harassment must be removed within a fixed time window, such as 24 hours after detection or user reporting. This encourages moderates to act as soon as possible, and this rule directly targets harmful content and provides a clear standard for acceptable moderation behavior.

A second rule is that the ranking algorithm must not amplify content that has been flagged as potentially harmful. For example, once a post is flagged or reported for misinformation or offensive content, it should not be recommended or promoted to additional users while it is under review. This ensures that the social media platform is not harming any users while it is flagged for certain harms. If the user deliberately searches for the post via a users profile, the rule could further state that there must be an apparent warning and flag on the post to ensure the user knows that the post may contain harmful information.

To provide a tamper-evident record that these rules are being followed, the system can generate detailed logs for each moderation decision. These logs would include the content identifier, the time it was flagged, the time action was taken, the type of action (such as removal or demotion), and the reason for the decision. Each log entry could be cryptographically signed or stored in an append-only system so that it cannot be modified without detection.

These records would allow auditors or regulators to verify that harmful content is consistently handled according to the stated rules, and that the platform is not selectively enforcing its policies.

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## B.2

Technical accountability alone is unlikely to fully satisfy public demand for more effective content moderation. While tamper-evident records can show whether a system is following its rules, they do not guarantee that the rules themselves are appropriate or sufficient.

For example, a platform could strictly follow its own moderation rules while still allowing harmful content to spread if those rules are too weak or too narrowly defined. However, strict rules may prevent non-harmful and useful content from being seen, and this could be a breach of freedom of speech and seen as a social media platforming overreaching on its powers. In addition, many decisions in content moderation involve subjective judgment, such as determining what qualifies as misinformation or offensive speech. These decisions may not be fully captured by technical records, as offensive vs non-offensive posts is not black and white but a deep grey area..

Technical accountability is still valuable because it provides transparency and evidence that can support legal and political accountability. Regulators, courts, or the public can use this evidence to evaluate whether platforms are acting responsibly. However, broader oversight, including policy decisions, public input, and legal standards, is still necessary to ensure that content moderation effectively addresses social harms and ensures the platform is a safe space for all..